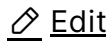


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Longitudinal modes of metallic nanostructures

This set of notes is on the paper [Nonlocal Response of Metallic Nanospheres Probed by Light, Electrons, and Atoms](https://pubs.acs.org/doi/10.1021/nn406153k)

<https://pubs.acs.org/doi/10.1021/nn406153k>, by **Christensen et al**

As done in the supplement of the paper, we seek longitudinal plasma solutions ($\mathbf{J}(\mathbf{r}, \omega) = \nabla \Psi(\mathbf{r}, \omega)$) in a metallic sphere. We note that such longitudinal modes must satisfy the equation (using the notation of the paper):

$$(\nabla^2 + k_{NL}^2)(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) = 0 \rightarrow (\nabla^2 + k_{NL}^2)(\nabla^2 \Psi(\mathbf{r}, \omega)) = 0,$$

where k_{NL} is defined in the paper and will be re-derived below. By requiring that $\Psi(\mathbf{r}, \omega)$ is a solution to the scalar Helmholtz equation, we obtain $\Psi(\mathbf{r}, \omega) = j_l(k_{NL}r)Y_{lm}(\theta, \phi)$. Furthermore, since $\mathbf{e}_r \cdot \mathbf{J}(\mathbf{r}, \omega) \propto \partial_r j_l(k_{NL}r)$ (and since we require that the normal component of the current vanish at the sphere's boundary, $r = R$), we obtain the condition $j'_l(k_{NL}R) = 0$. Furthermore, note that k_{NL} may be readily found through Maxwell's equations combined with the hydrodynamic Ohm's law :

$$\begin{aligned} \frac{\beta_F^2}{\omega(\omega + i\eta)} \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \mathbf{J}(\mathbf{r}, \omega) &= \frac{i\omega_p^2 \epsilon_0}{\omega + i\eta} \mathbf{E}(\mathbf{r}, \omega) \rightarrow \\ \frac{\beta_F^2}{\omega(\omega + i\eta)} \nabla^2(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \nabla \cdot \mathbf{J}(\mathbf{r}, \omega) - \frac{i\omega_p^2 \epsilon_0}{\omega + i\eta} \nabla \cdot \mathbf{E}(\mathbf{r}, \omega) &= 0 \rightarrow \\ \frac{\beta_F^2}{\omega(\omega + i\eta)} \nabla^2(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \left[1 - \frac{\omega_p^2}{\omega(\omega + i\eta)} \right] \nabla \cdot \mathbf{J}(\mathbf{r}, \omega) &= 0, \end{aligned}$$

where β_F has velocity units (and is typically taken as $\sqrt{3/5}v_F$ for a three dimensional metal and where we used the following (which follows from Ampere's law):

$$0 = \nabla \cdot (\nabla \times \mathbf{H}(\mathbf{r}, \omega)) = -i\omega\epsilon_0 \nabla \cdot \mathbf{E}(\mathbf{r}, \omega) + \nabla \cdot \mathbf{J}(\mathbf{r}, \omega) \rightarrow \\ \nabla \cdot \mathbf{E}(\mathbf{r}, \omega) = (-i/\omega\epsilon_0) \nabla \cdot \mathbf{J}(\mathbf{r}, \omega)$$

Thus, we have:

$$k_{NL}^2 = \frac{1}{\beta_F^2} \left[\omega(\omega + i\eta) - \omega_p^2 \right]$$

Therefore, for purely longitudinal resonances, we obtain the following dispersion relation for a given zero of the spherical Bessel function's derivative, w_v :

$$k_{NL}^2 = w_v^2/R^2 \rightarrow \omega(\omega + i\eta) = \omega_p^2 + \frac{\beta_F^2 w_v^2}{R^2}$$

This is **Equation S22** of the supplement of the paper as well **Equation 9** of the main text.